

Shenglei Fang

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EDUCATION BACKGROUND

2022.09–	Cardiff University	PhD Student
● Deep learning, Medical image, Multi-Modality, long sequence, single cell transcriptomics		
2019.09–2021.03	Cardiff University	Data Science and Analytics (MSc)
● Core courses: Computational Data Science, Applied Machine Learning , Supply Chain Modelling etc		
2013.09–2017.06	Shanghai Maritime University	Information and Computing Science
● Core courses: Algorithm and Data Structure, Matlab Programming, Information theory, Data Analysis,etc		

HONORS & AWARDS

- 2nd class scholarship of Shanghai Maritime University in 2017(Top 10%)
- 3rd class scholarship of Shanghai Maritime University in 2016
- Cardiff University WM Thomas Fund £650
- 2024年Cardiff University I&I annual Meeting Poster Presentation
- 2025年CardiffUniversity I&I annual Meetion Oral Presentation
- ISBM/ECCB 2025 Poster Presentation
- ICML 2026 Poster Presentation(Top 15%)

RESEARCH EXPERIENCE

2025.07–2026.01 **VBA: Vector Bundle Attention for Intrinsically Geometric Representation Learning (First Author, ICML 2026)**

Developed Vector Bundle Attention (VBA), a geometry-informed Transformer architecture that models tokens as base – fiber pairs with learned transport and curvature-aware attention. Validated across single-cell transcriptomics, spatial transcriptomics, and 3D point cloud classification, achieving strong performance in cell annotation, tissue domain clustering, and ModelNet40 recognition.

2025.11–2026.01 **Dynamic Fractal Mamba: A Neural Renormalization Group Flow for Scale-Invariant Sequence Modeling (First Author, ICML 2026)**

Developed DF-Mamba, a fractal geometry-inspired multi-scale state space model for robust cross-scale representation learning, validated across time-series forecasting, spatial transcriptomics clustering, and computational pathology classification. Achieved strong zero-shot scale generalization by sharing recursive dynamics across resolutions, improving robustness on $\times 2/\times 4$ coarsened inputs while maintaining efficient FLOP usage.

2022.10–2026.05 **Development of a Deep Learning Model to Integrate Diagnostics for Noninvasive Detection of Fatty Liver Disease (First Author, Under Review)**

Developed multimodal deep learning models (DeepLUS+) for noninvasive MASH diagnosis, validated on multi-center biopsy datasets and >7,000 real-world cases. Achieved AUC >0.96, outperformed clinicians, and deployed an offline demo system with strong population-level consistency in diabetic risk prediction.

2024.07-2025.05

scSEEK: a Self-supervised Expert Embedding foundation model with Reinforcement-tuned Knowledge for Single-cell RNA-seq (First Author, Under Review)

Developed scSEEK, the first self-supervised framework integrating expert-aware Transformers and reinforcement learning for single-cell RNA-seq analysis. Introduced a biologically-informed Mixture-of-Experts (Bio-MoE), Multi-Head Latent Attention (MLA), and a group-aware RL strategy (Bio-GRPO). Trained on 599k cells, scSEEK achieved SOTA performance on PBMC, liver, and pancreas datasets, surpassing scBERT/scGPT. Demonstrated strong generalization, novel cell discovery, and robust clustering (ARI=0.977).

2024.12-2025.03

A Universal Hybrid Transformer-Mamba Model across multiple imaging modalities for Disease Diagnosis (First Author, Under Review)

Developed a hybrid deep learning model combining Gated Attention-enhanced Vision Transformers and the Mamba state-space architecture for multi-modality medical image classification. Achieved high accuracy across tasks: 92.3% (liver fibrosis), 93% (pneumonia), 85% (breast cancer), 83.9% (skin cancer), and 81.8% (Alzheimer' s). Demonstrated strong robustness and generalization on histopathology, MRI, X-ray, and clinical image datasets

2020.12-2024.08

Prediction of Alzheimer's disease based on Biomarkers and images (First Author, Under Review)

On the basis of the master thesis, MRI images and biomarkers in ADNI were added, and convolutional neural network model was used to establish classification and prediction models. Processed 150G of image data and more than 1 million medical data, increasing the accuracy of the previous model to about 97%, which is a huge improvement over the model of a year ago.

WORK EXPERIENCE

2021.09 - 2022.10	Beijing 58 Information Technology Co., Ltd	Data Scientist
2020.10-2021.09	Shanghai Zhang Xiao Meng Education Technology Co., Ltd	Data Analyst
2017.09-2019.06	Shanghai Zhang Xiao Meng Education Technology Co., Ltd	Data Analyst

OTHER INFROMATION

- IT Skills: Python, SPSS, R, Matlab , SQL , C/C++
- As Reviewer of ICML2026 (**Silver Reviewer**) , Neurips2026 and AAAI 2027